AI–Assisted Learning and Research:
Production, Verification, and Tool Integration

Carlos Mendez

Nagoya University, Japan

carlos-mendez.org

Agenda

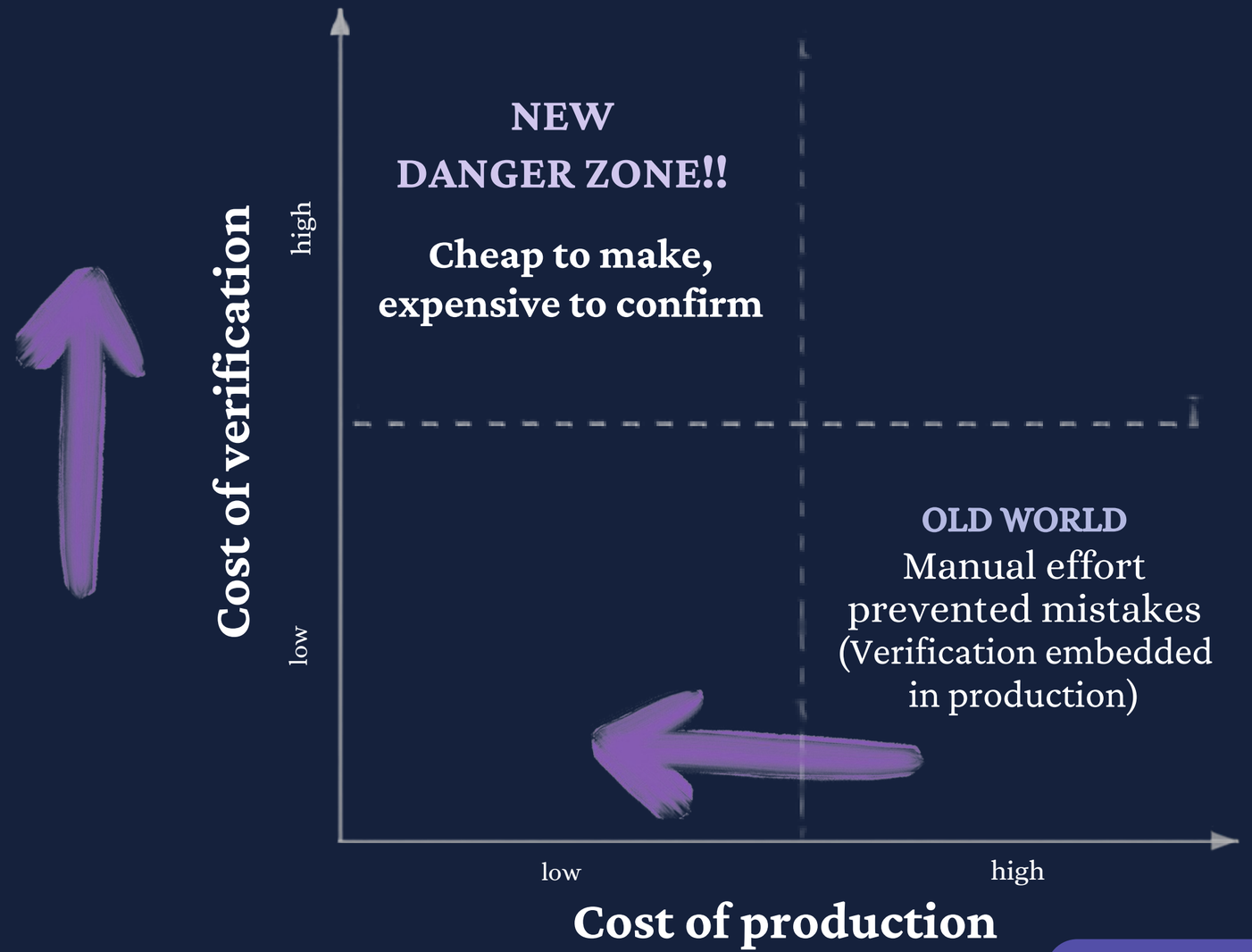
1. A conceptual framework
2. Example 1: AI-assisted learning
3. Example 2: AI-assisted research

Section 1

A Conceptual Framework:

The agentic AI trade-off

Lower production costs with...



... higher verification costs

Source: Adapted from Clifford Russel (https://substack.com/home/post/p-203010751)

Section 2

Example 1: AI-Assisted Learning



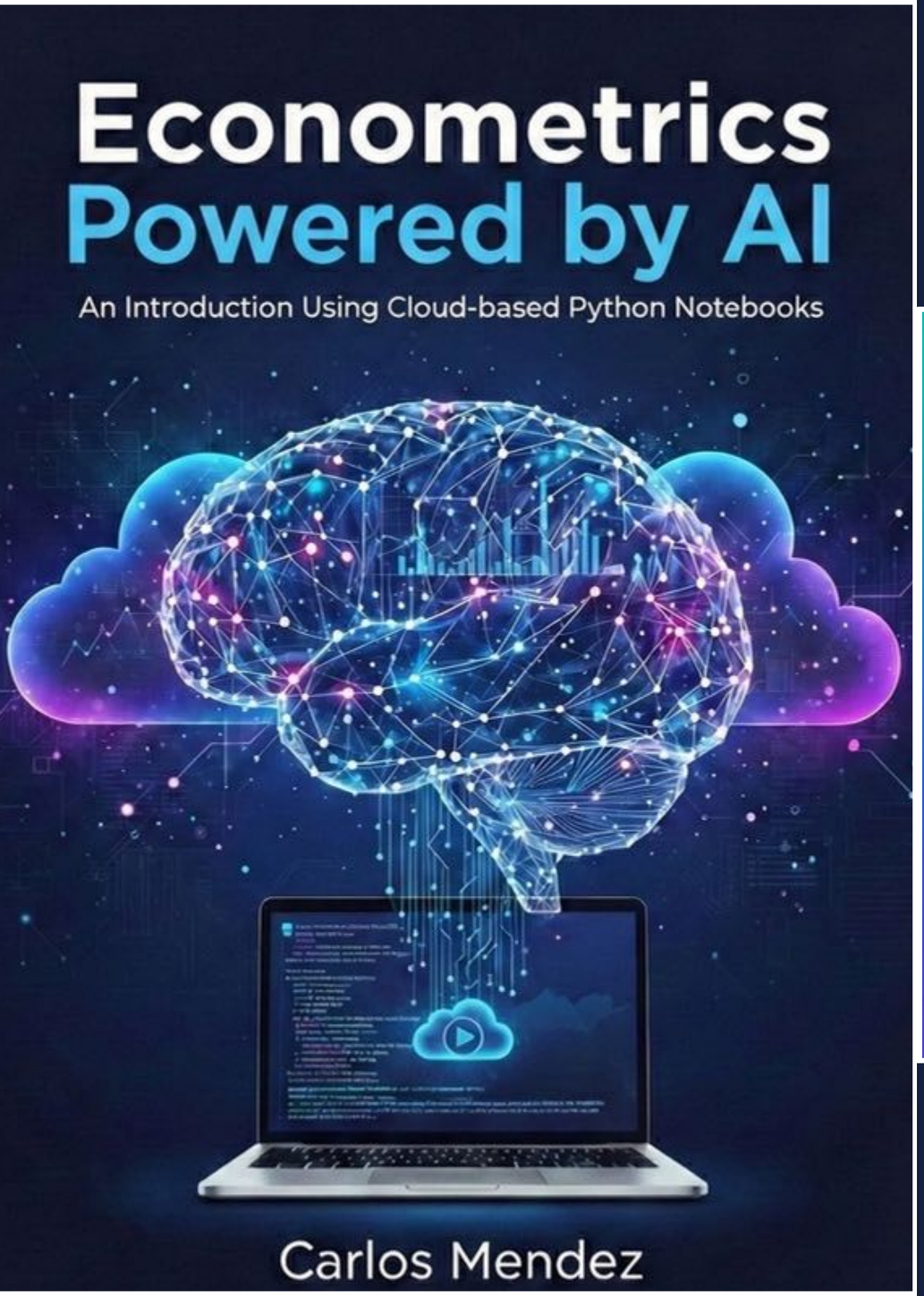
Challenges of teaching/learning econometrics in graduate programs in public policy and development studies


Learning econometrics should
be
a more **accessible** and
interactive experience

Python in the Google Colab cloud provides both...



...access and interactivity

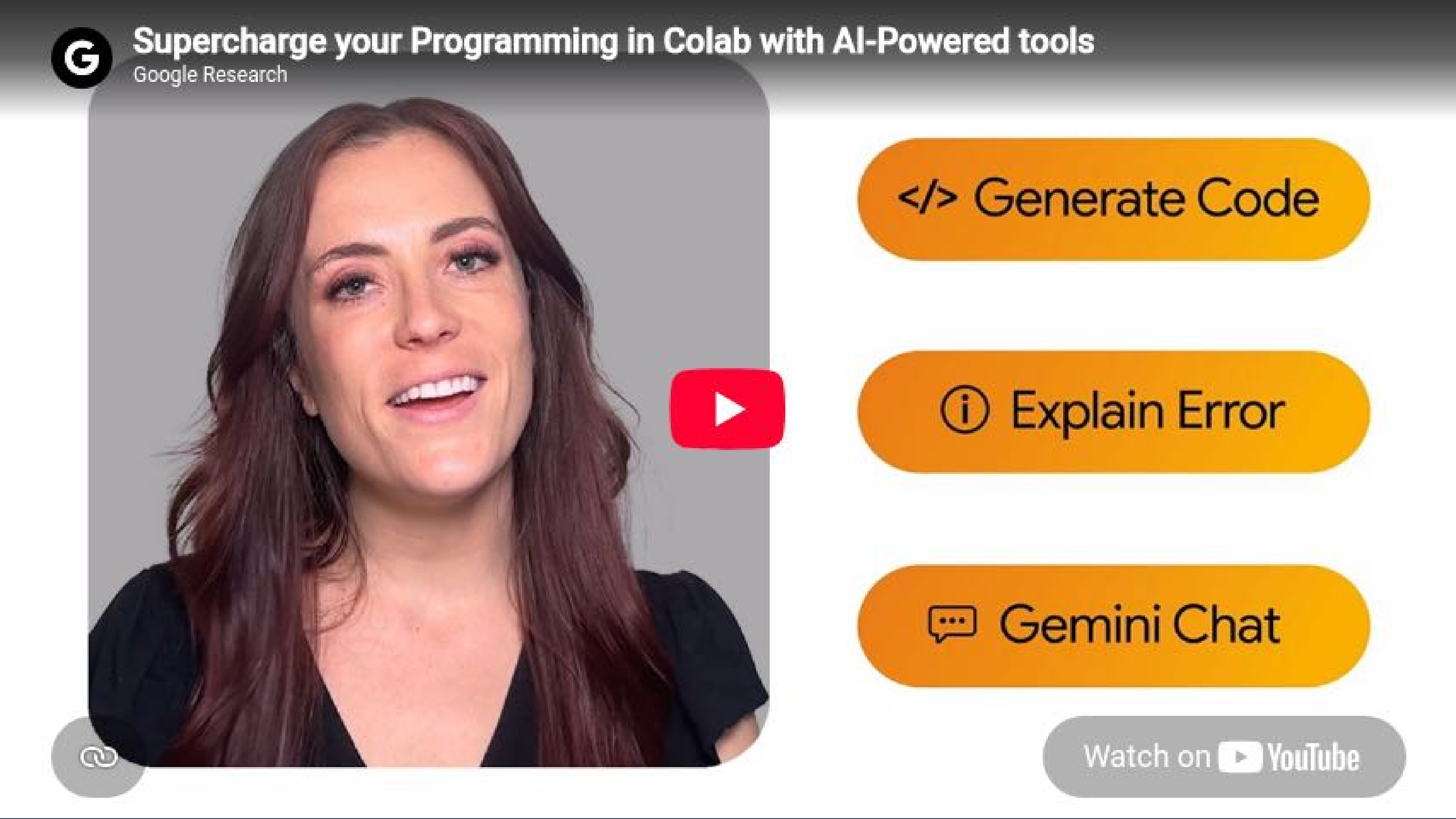


metricsAI

An Introduction to Econometrics with Python and AI in the Cloud



https://quarcs-lab.github.io/metricsai/


Supercharge your Programming in Colab with AI-Powered tools
Google Research
</> Generate Code
Explain Error
Gemini Chat
Watch on YouTube

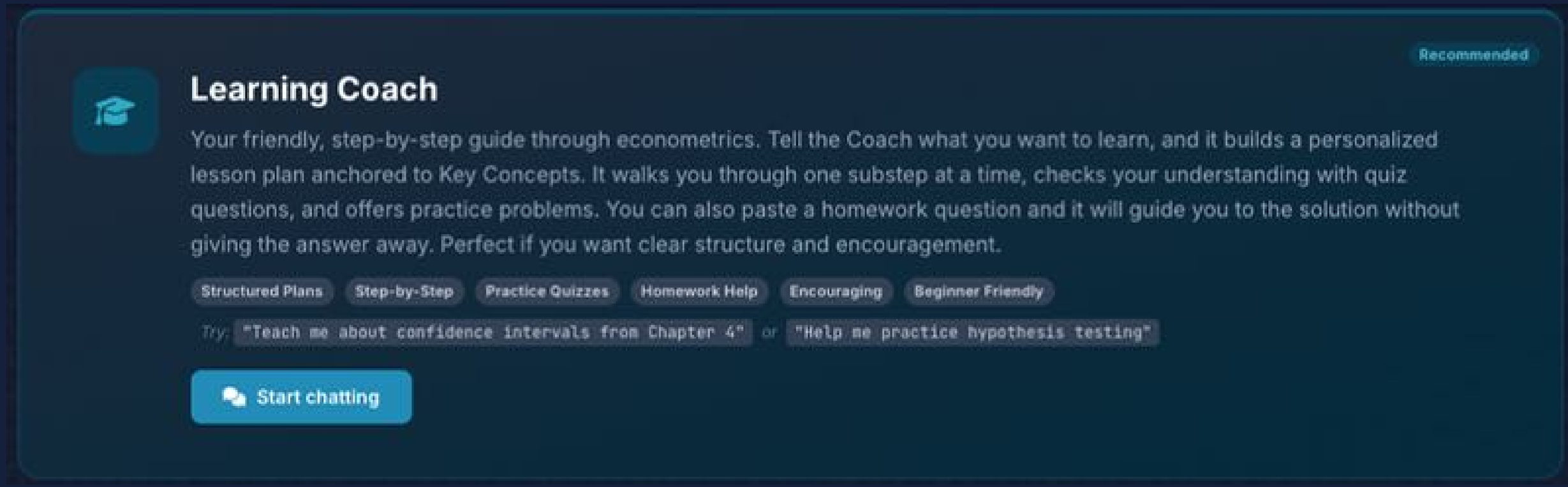
Other AI-products

Diverse collection of AI tutors...



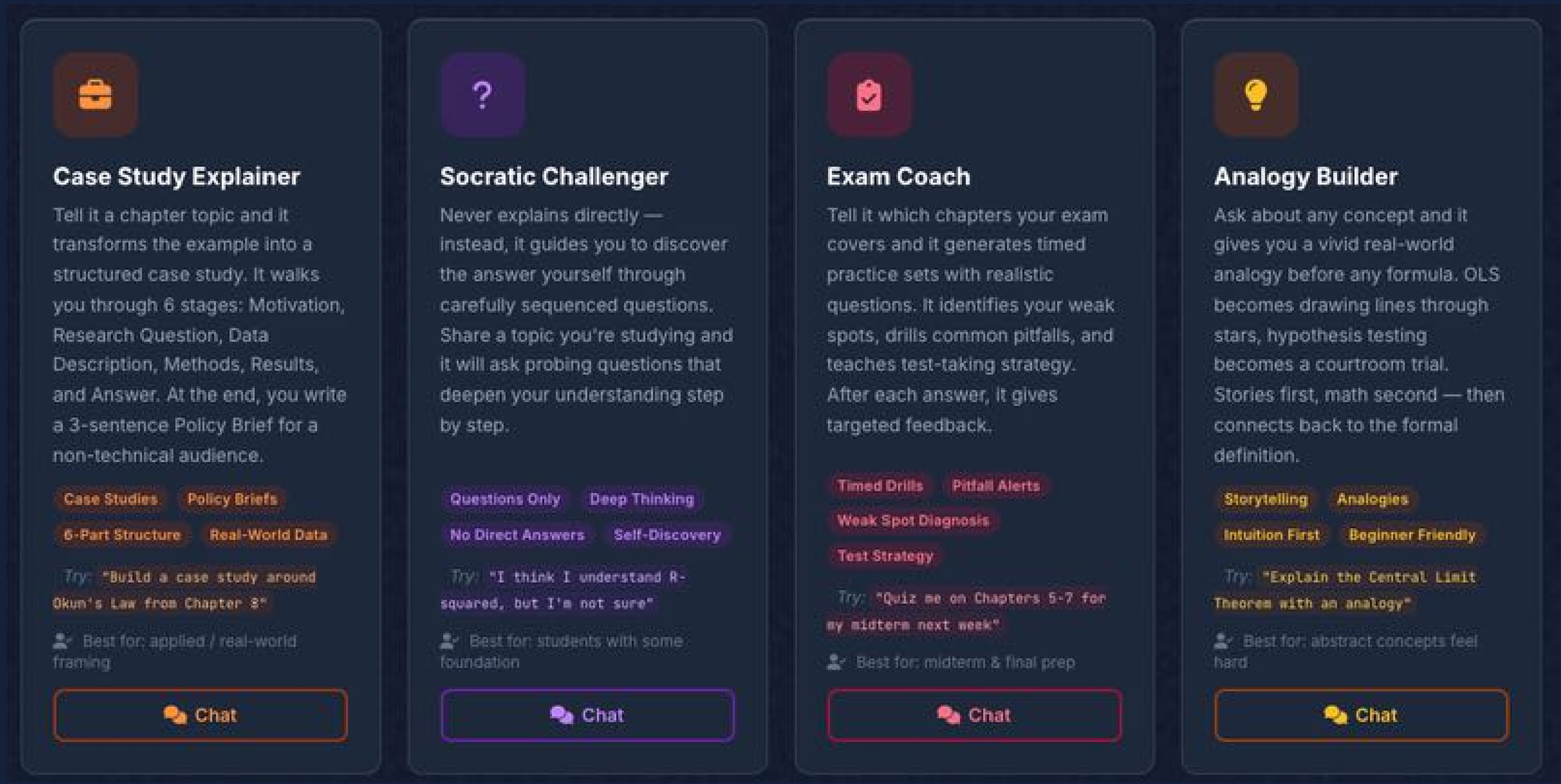
... to personalize your learning and foster critical thinking

Diverse collection of AI tutors...



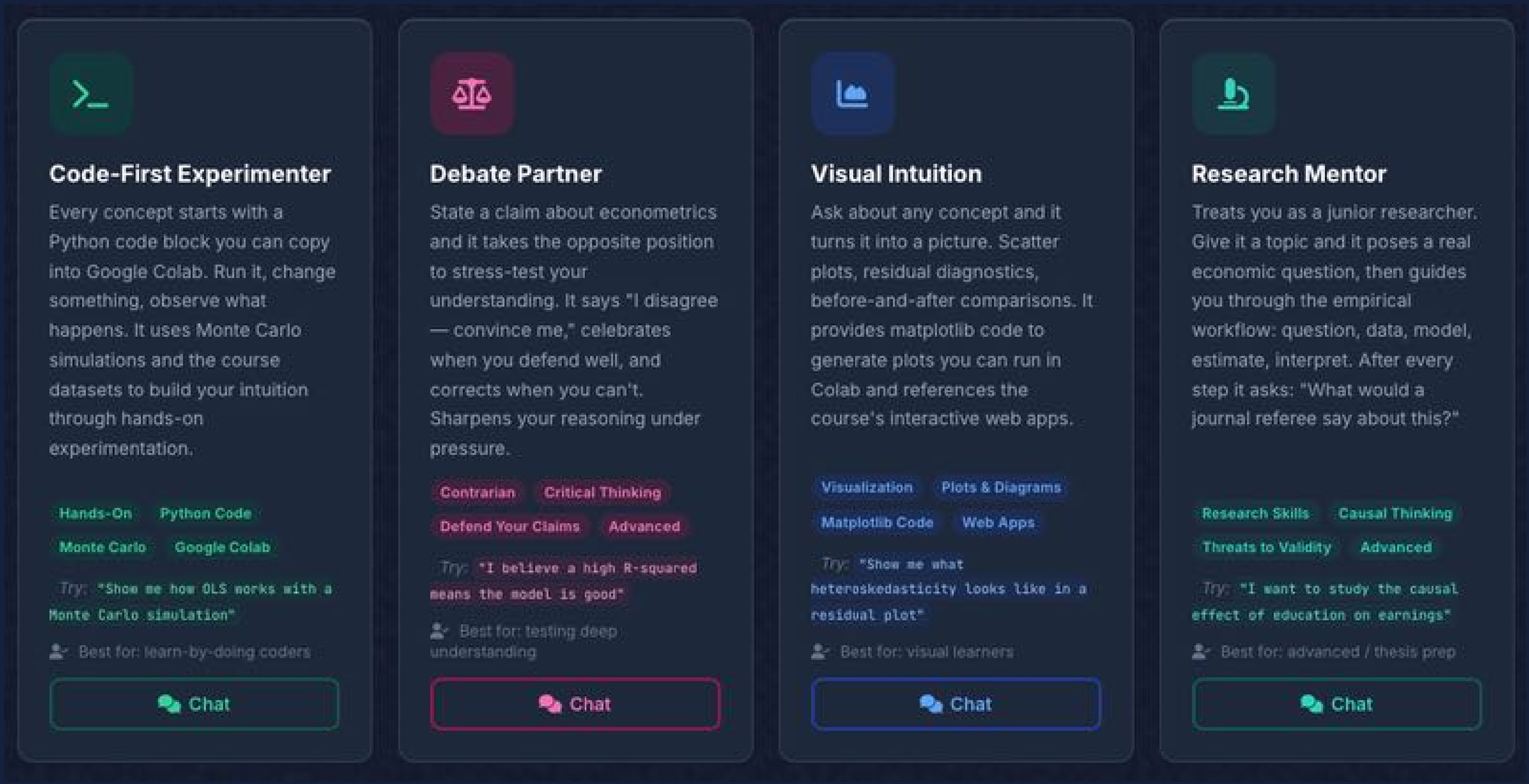
... to personalize your learning and foster critical thinking

Diverse collection of AI tutors...



... to personalize your learning and foster critical thinking

Diverse collection of AI tutors...



... to personalize your learning and foster critical thinking

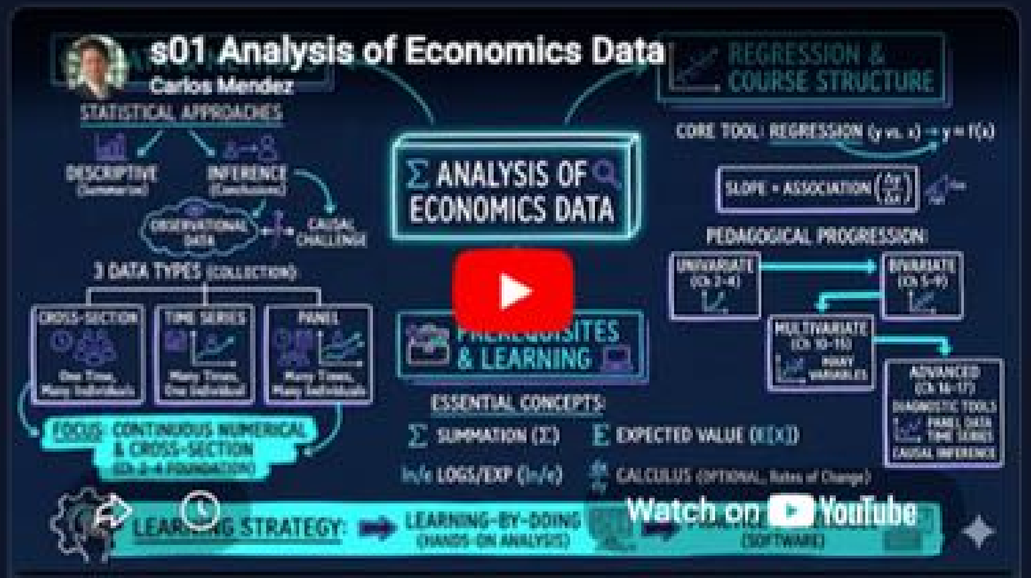
17 AI Video Lectures

Short Video Overviews

AI-generated video lectures for every chapter. Watch them here or on YouTube.

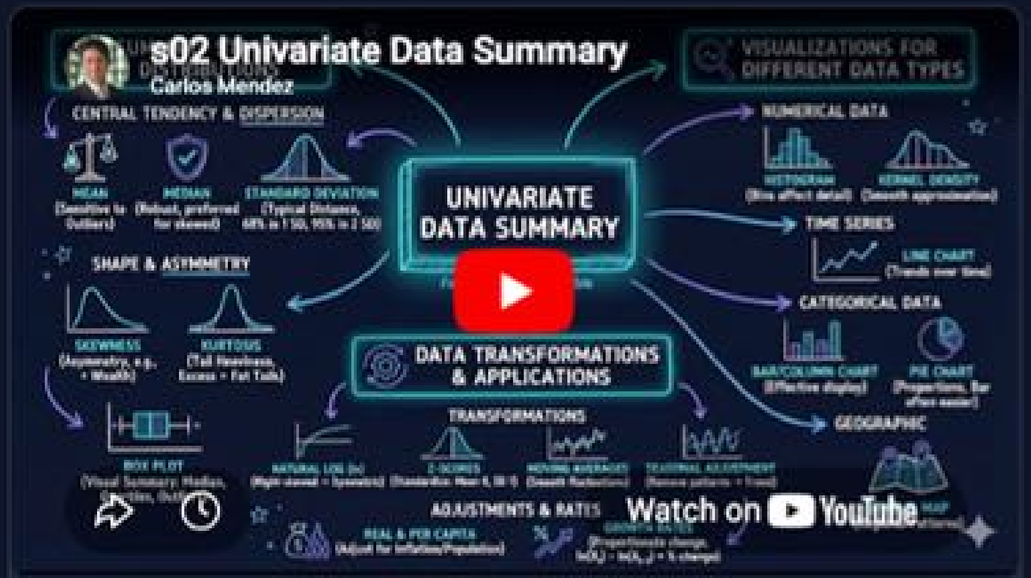
Watch on YouTube

1 Part I: Statistical Foundations



Chapter 1: Analysis of Economics Data

Regression basics, scatter plots, OLS, R-squared, association vs. causation



Chapter 2: Univariate Data Summary

Histograms, box plots, summary statistics, log transformations, z-scores



s01 Analysis of Economics Data

Carlos Mendez

DATA & METHODS

STATISTICAL APPROACHES

- DESCRIPTIVE (Summarize)
- INFERENCE (Conclusions)
- OBSERVATIONAL DATA ↔ CAUSAL CHALLENGE

3 DATA TYPES (COLLECTION):

- CROSS-SECTION: One Time, Many Individuals
- TIME SERIES: Many Times, One Individual
- PANEL: Many Times, Many Individuals

FOCUS: CONTINUOUS NUMERICAL & CROSS-SECTION (Ch 2-4 FOUNDATION)

REGRESSION & COURSE STRUCTURE

CORE TOOL: REGRESSION (y vs. x) → $y \approx f(x)$

SLOPE = ASSOCIATION $\left(\frac{\Delta y}{\Delta x}\right)$ (rise / run)

PEDAGOGICAL PROGRESSION:

- UNIVARIATE (Ch 2-4)
- BIVARIATE (Ch 5-9)
- MULTIVARIATE (Ch 10-15): MANY VARIABLES
- ADVANCED (Ch 16-17): DIAGNOSTIC TOOLS; PANEL DATA; TIME SERIES; CAUSAL INFERENCE

Σ ANALYSIS OF ECONOMICS DATA

PREREQUISITES & LEARNING

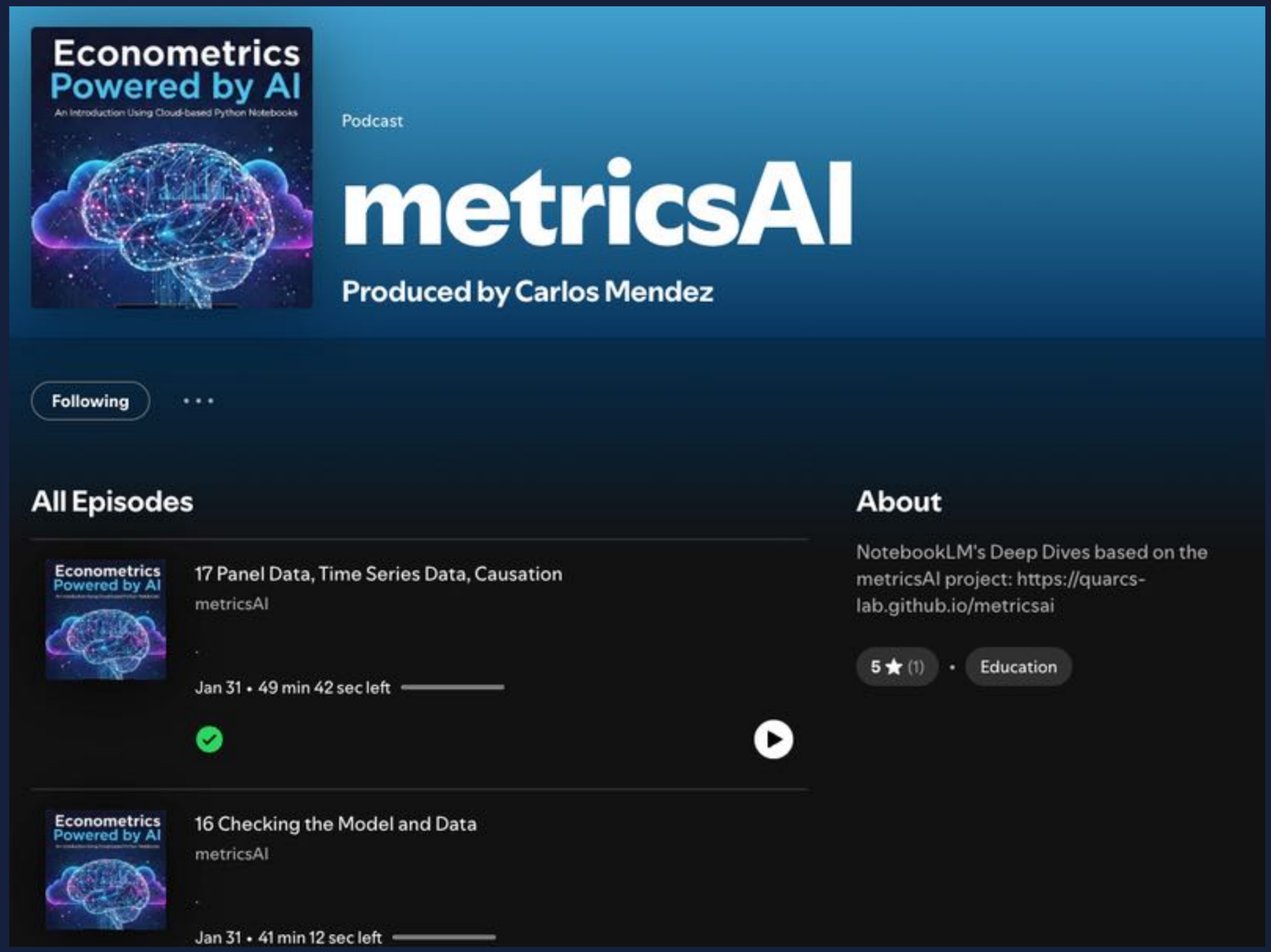
ESSENTIAL CONCEPTS:

- Σ SUMMATION (Σ)
- 𝔼 EXPECTED VALUE (𝔼[X])
- ln/e LOGS/EXP (ln/e)
- $\frac{dx}{dy}$ CALCULUS (OPTIONAL, Rates of Change)

LEARNING STRATEGY: → LEARNING-BY-DOING (HANDS-ON ANALYSIS) → PRACTICE & EXERCISES (SOFTWARE)

Watch on YouTube

AI Podcasts Series ...



... accesible via Spotify

Key lesson:

From “prompt engineering” to “designing AI agent skills”. Use AI agent skills to generate consistent outputs, define sequential tasks, and apply verification tests

Hint: Help me design an agentic skill about <YOUR TASK HERE>


NEW
Ultimate Guide To Claude Skills
Skill Leap AI
CLAUDE
SKILLS
SKILL LEAP AI
HubSpot Media
Watch on YouTube

A workflow example:

Activate planning mode

Type your high-quality prompt. Ideally use agentic skills that include reference materials, examples, and verification mechanims

Before you start planning, ask me questions until you are 95% confident you understand exactly what I need. Do not make any assumptions.

Section 3

Example 2: AI-Assisted Research

The Prompt

```
Estimate abnormal returns after IPOs
using Jay Ritter's data and CRSP.
```

The Result

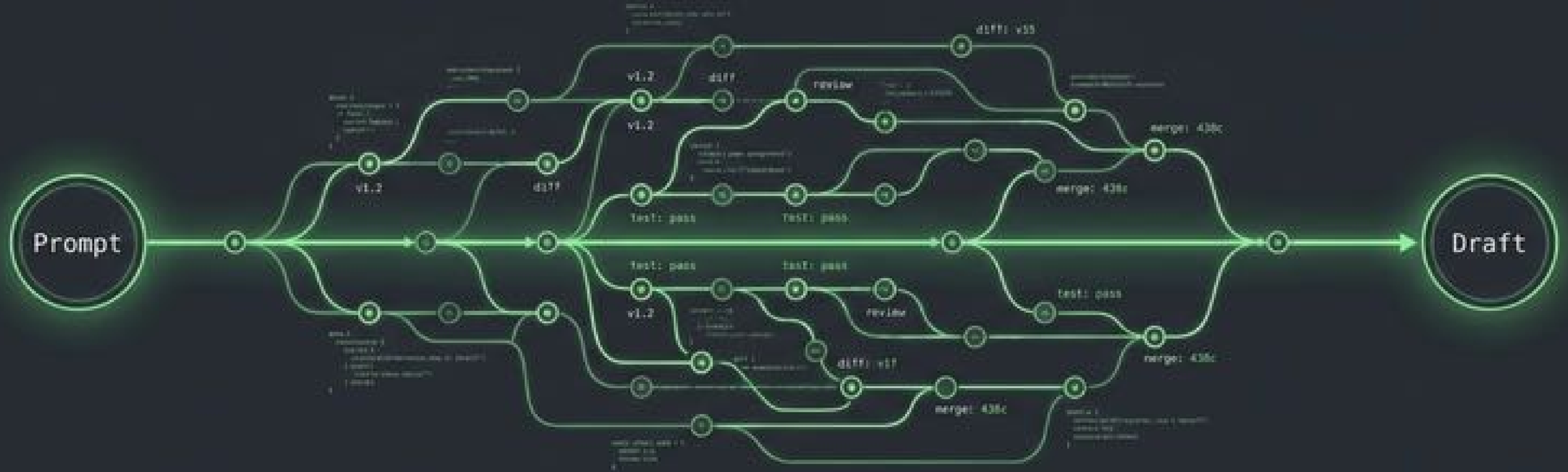


Looks perfect. Passes the sniff test.
But how do we prove it's right?

Source: Paul Goldsmith-Pinkham (https://paulgp.substack.com/p/integration-and-collaboration-in1).

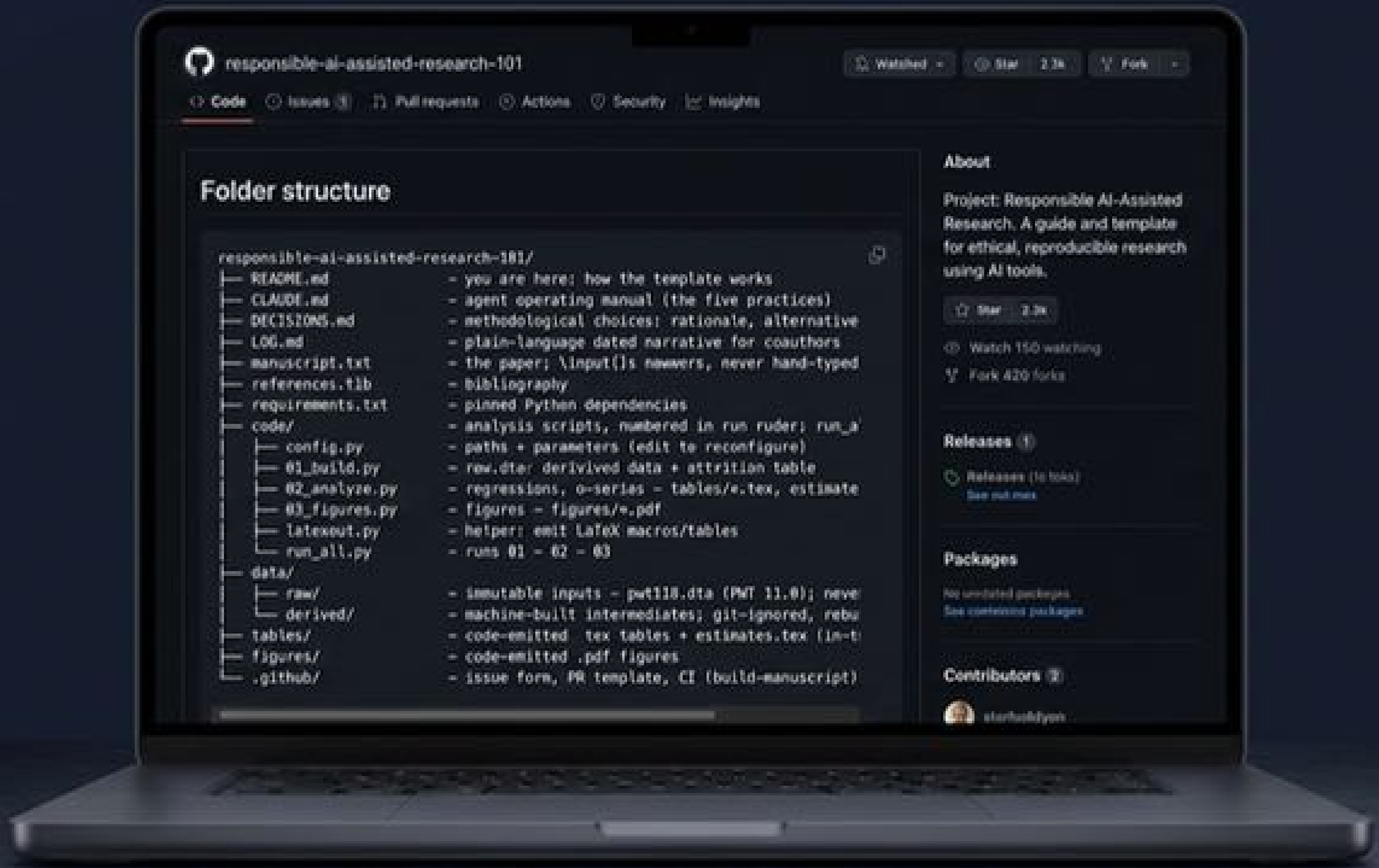
The Transparent Pipeline

Overcoming verification debt in AI-assisted empirical research.



... efficiently and transparently audit the decisions of your AI agent

Claude Code + Github + Overleaf



https://github.com/quarcs-lab/responsible-ai-assisted-research-101

The Prompt

```
Estimate abnormal returns after IPOs
using Jay Ritter's data and CRSP.

CRITICAL: Commit code locally at every
meaningful checkpoint. Keep a
DECISIONS.md and LOG.md.
```

The Result



Looks perfect. Passes the sniff test.
But how do we prove it's right?

Source: Paul Goldsmith-Pinkham (https://paulgp.substack.com/p/integration-and-collaboration-in1).

Changing the unit of verification...

Cognitive Overload (800 lines)



Bounded Diff (12 lines)

```
@@ -1,3 +1,12 @@
 dtera_measures = sanitAng.decention, m.month(2)
 dtara = co.dinct(1)
 
+# fixed-horizon monthly abnormal return for the CAR analysis: while the firm trades
+firma = anavst.return.iarrose()
+while firm in firm = ann.confirm_chaseClonathis):
+    firm = m data.dine.exists(
+    if (teams.returnassetresult.renotfolite return.monthis) {
+        antens.returnareturns.hrisantual()
+        return.retrieassstenath.return(stavse))
+    }
 # Unite DIM debug to traderare
 return tesent
```

Validate constructed measures against Ritter's series (stage 7, closes #1) `83175a7`
paulgp and claude committed last week

Add cumulative abnormal return (CAR) analysis (stage 6) `84aaa6f`
paulgp and claude committed last week

Add results stage, entry points, README, and full logs `4e97f5a`
paulgp and claude committed last week

Add stages 3-4: CRSP pull and BHAR/CAR computation `88894db`
paulgp and claude committed last week

Add stages 1-2: load Ritter events and validate against CRSP `8893fcf`
paulgp and claude committed last week

Scaffold IPO long-run abnormal-return project `6e74e73`
paulgp and claude committed last week

...you review a story step-by-step, not a monolithic block of text.

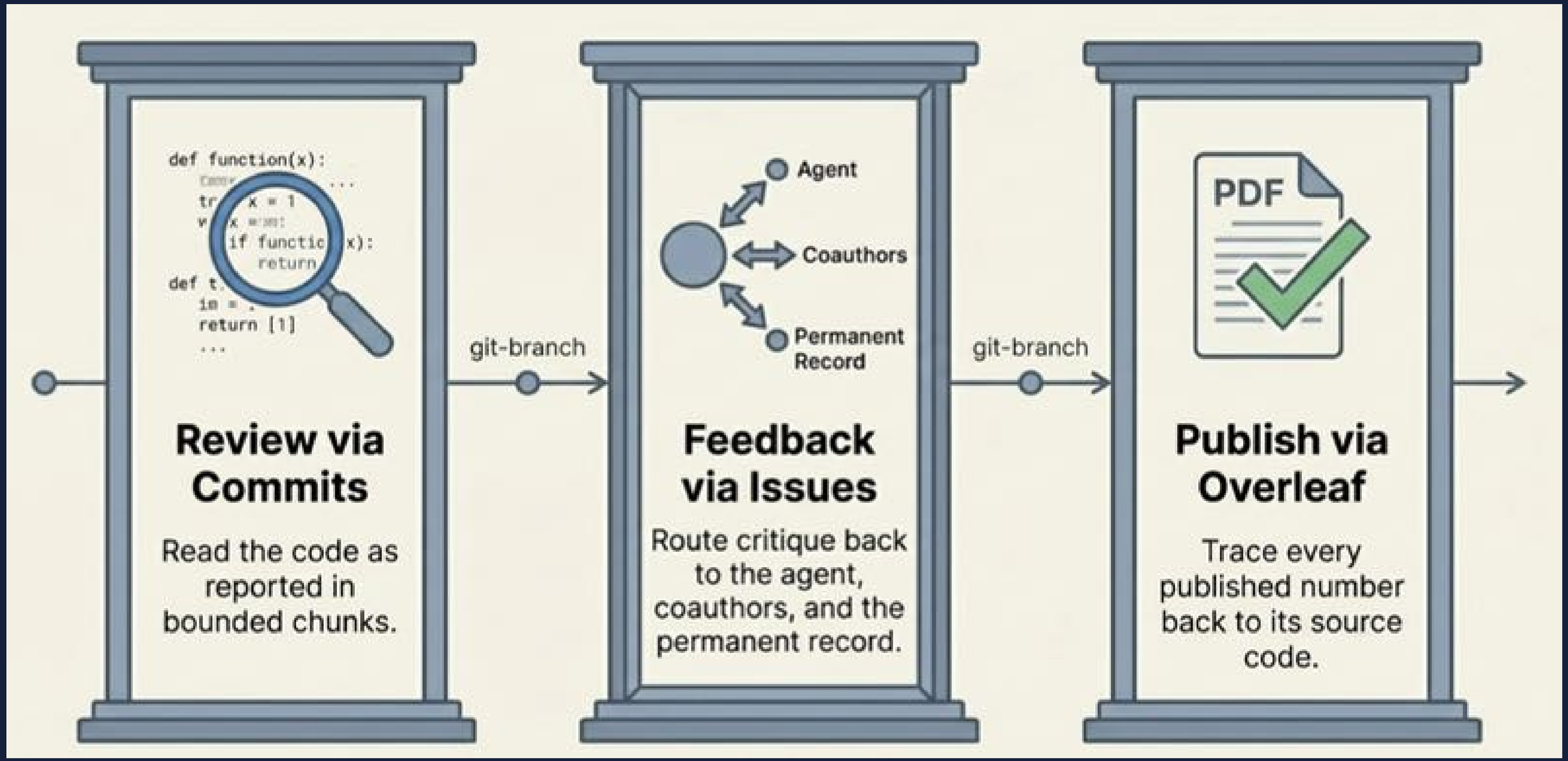
GitHub helps you ...



... manage the discussion of the code, results, and story of the manuscript

... avoid black-box research

A work environment built for verification ...



Source: Paul Goldsmith-Pinkham (https://paulgp.substack.com/p/integration-and-collaboration-in1).

In the age of AI, Git and Github are easier to use...

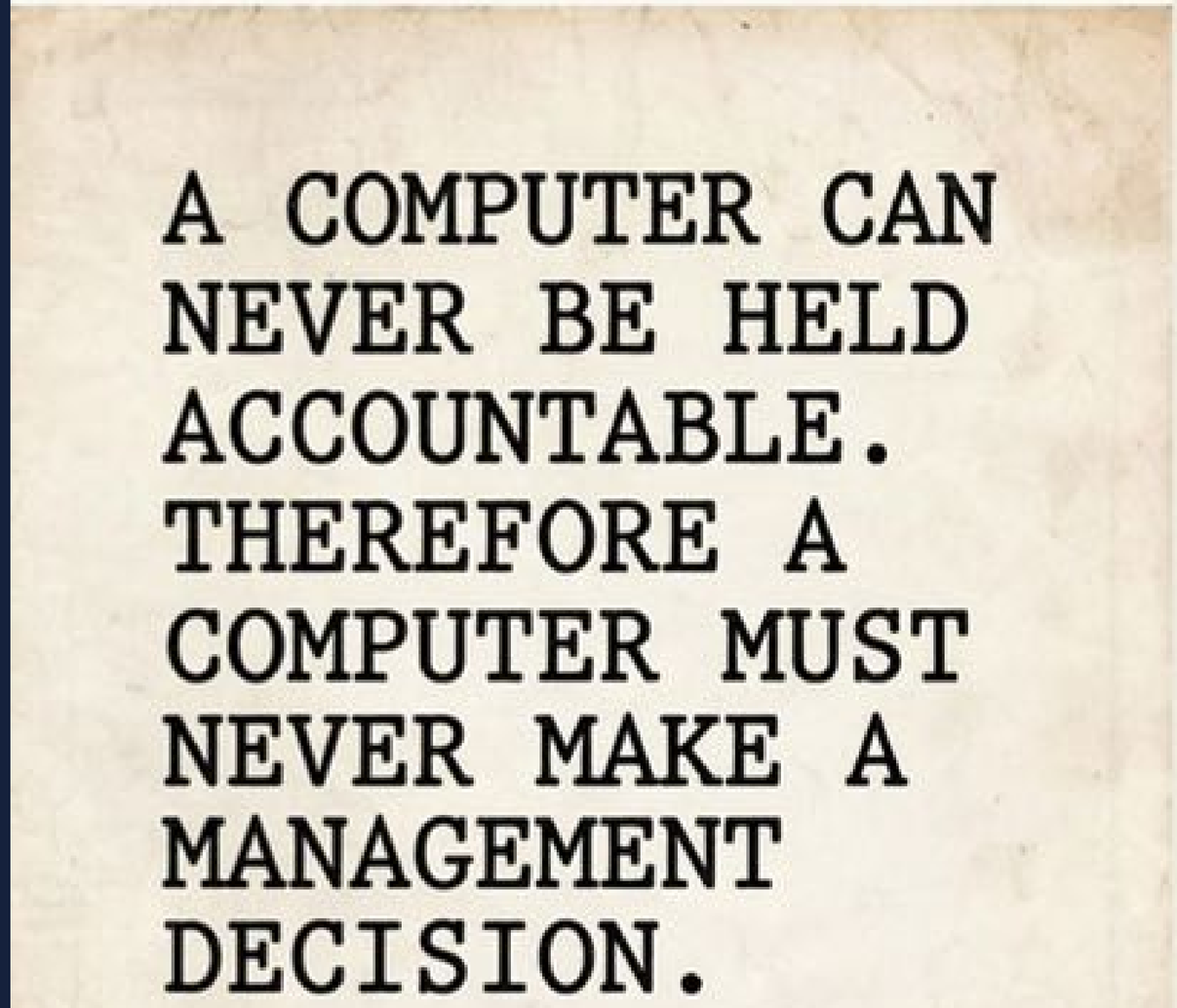
Researchers avoided version control because the setup cost was too high. Today, that cost is zero.

```
Prompt: Please track everything we do in git and GitHub.
```

Claude has the spirit of a senior software engineer built in. You get enterprise-grade best practices without accumulating the years of scar tissue required to learn them.

...your AI agent can teach you how to use GitHub effectively

The Accountability Imperative



Source: 1970s IBM training manual

1. Gemini can review Claude's code to catch syntax errors.
2. But methodological sign-off is a human constraint.
3. **This workflow doesn't replace the researcher; it scales their ability to verify.**


Integration & Collaboration: Claude Code for Econ. w/ P. Goldsmith-Pinkham | Mark
Markus' Academy
Claude Code for Applied Economists
Episode 8: Integration and Collaboration
markus'academy
July 3, 2026
Paul Goldsmith-Pinkham
PRINCETON UNIVERSITY BENDHEIM CENTER FOR FINANCE
PrincetonEconomics
Watch on YouTube

Concluding remarks

1. Understand the AI trade-off: Production and verification costs
2. AI can assist both learning and research, but verification is needed
3. Integrating agentic AI skills and GitHub tools can be helpful for both production and verification

Thank You

Scan to access these slides

Carlos Mendez
Nagoya University, Japan
carlos-mendez.org